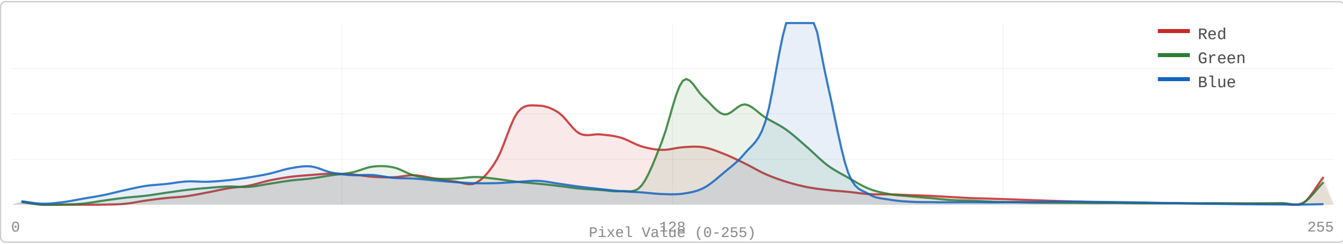
Decorrelation Gap: KODIM21

768x512 | 24-bit RGB | PNG (lossless)

Village Houses and Mountains | Weakly one-dimensional (PC1 = 88.7%)

1. RGB Channel Distribution



2. Inter-Channel Correlation: RGB vs YCbCr

RGB (Original)

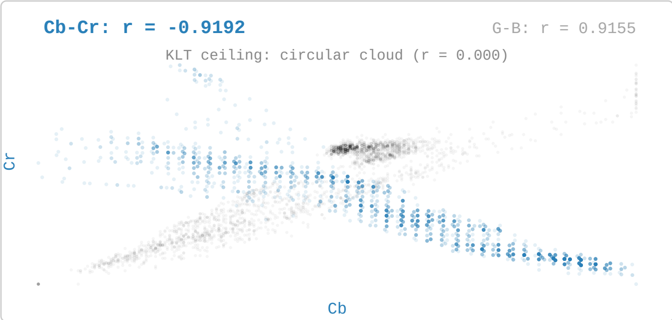
	R	G	B
R	1.0000	0.8957	0.6651
G	0.8957	1.0000	0.9155
B	0.6651	0.9155	1.0000

Cb-Cr:
-0.919
(KLT ceiling = 0.000)

YCbCr (BT.601)

	Y	Cb	Cr
Y	1.0000	0.0018	-0.2371
Cb	0.0018	1.0000	-0.9192
Cr	-0.2371	-0.9192	1.0000

3. Cb vs Cr Scatter with G-B background overlay



4. Pairwise Decorrelation Gap

Channel Pair	RGB	YCbCr (BT.601)	Residual |r|
R-G → Y-Cb	0.895717	0.001766	0.0018
R-B → Y-Cr	0.665054	-0.237086	0.2371
G-B → Cb-Cr	0.915487	-0.919215	0.9192
Avg |r|	0.825419	0.386023	0.3860

Decorrelation Gap: KODIM21

768x512 | 24-bit RGB | PNG (lossless)

Village Houses and Mountains | Weakly one-dimensional (PC1 = 88.7%)

5. Decorrelation Efficiency & Blue Independence

Decorrelation Efficiency

53.2% Moderate

0% BT.601 leaves all — 100% KLT ceiling

Blue Channel Independence (Baetzel 2026a)

13.0%

0% Fully predictable — 100% Fully independent

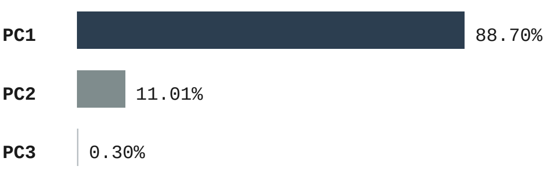
6. 4:2:0 Chroma Subsampling — Information Cost

Channel	PSNR (dB)
R	45.92 dB
G	49.95 dB
B	42.01 dB
Overall	44.83 dB

Method: BT.601 RGB→YCbCr | 2x box downsample Cb,Cr | nearest upsample | BT.601 inverse

7. PCA Baseline Reference (Baetzel 2026a)

Eigendecomposition



Eigenvector Loadings

	R	G	B	Eigenvalue
PC1	-0.5061	-0.6078	-0.6119	5346.89
PC2	0.7421	0.0545	-0.6680	663.52
PC3	-0.4394	0.7922	-0.4235	17.92

8. PCA Derived Metrics

Condition Number

298.45

PC1/PC2: 8.06 | PC2/PC3: 37.04

Blue Channel Independence

13.0%

Variance not captured by PC1

PC1 Dominant Channel

B

Loading: B (0.61)

9. Redundancy Profile

Classification: Weakly one-dimensional | CN: 298.45 | PC1: 88.7%

RGB Correlation: Avg |r| = 0.825419 | R-G: 0.8957 | R-B: 0.6651 | G-B: 0.9155

YCbCr Residual: Avg |r| = 0.386023 | Y-Cb: 0.0018 | Y-Cr: -0.2371 | Cb-Cr: -0.9192

Efficiency: 53.2% of KLT ceiling | Gap: 0.3860 avg |r| remaining

Blue Channel: 13.0% independent | 4:2:0 PSNR: 44.83 dB

References

[1] Baetzel, J. (2026a). "Statistical Characterization of Inter-Channel Redundancy Structure"
[2] Wallace, G.K. "The JPEG still picture compression standard." IEEE Trans. CE 38(1), 1992.
[3] ITU-R Recommendation BT.601: Studio Encoding Parameters, 1982.
[4] Watanabe, S. "Karhunen-Loeve Expansion and Factor Analysis," pp. 635-660, 1965.